M1 Working Note: The Modularization Campaign

Working note v0.1, Project secret3, Row A infrastructure, August 11, 2026

What was attempted and what stands

Line M1 asked for the monolith, Proofs.idr at 3,787 lines, to be split into an Idris2 package so that verification stops being all or nothing. The campaign delivered a package of eighteen modules with a battle-tested generator, and it closed with a verified foundation: **ten modules, Core through Quote plus the extracted Values module, build from scratch in under two minutes with zero errors**, covering the monolith's first 2,050 lines and the Val datatype, which is everything the valence and nucleic parts consume from part S. The upper region, parts T through AC, remains verified by the monolith route, full context to line 3,644 with parts AB and AC slice-verified, exactly as before; its module sources are generated and shipped for future work. The package, generator state, and build log are in proofs-package-m1-v0.1.tar.gz.

The generator and its five lessons

The splitter regen.py had to learn five real failure classes, each now a checker discipline. One: visibility must be injected, since a monolith needs none; 542 symbols were classified. Two: the monolith's own namespace qualifications, Proofs.iC and kin, written to defeat implicit auto-binding, must be requalified per symbol to each name's new home module. Three: comma-grouped signatures, iC, bC, ki, sb, skb : Term, must be parsed as declaring every name in the group. Four: equation-only constants with no signature must be found and exported, or downstream modules see nothing. Five: mutual-block members take their visibility inside the block, at the block's indentation.

The decisive design is **two-tier visibility**: public export only for what downstream proofs must unfold, data declarations, defining constants, computational and type-level functions, plus an error-driven whitelist mined from constraint failures; everything else, meaning proof bodies, gets plain export, because a proof is an opaque certificate. Blanket public export bloats every import interface and was the campaign's original sin.

The two anomalies, documented as findings

Finding one, the part S blowup. Part S, strong normalization by reducibility candidates, elaborates inside the monolith within the 4 GB sandbox, historical fact. The same lines, byte-identical, as a module behind imports, exceed 12 GB and die; the TypedX experiment proved the body unmodified. Bisection localizes the explosion to the redSxyzGo mutual block, lines 2177 to 2219. Neither minimal imports, nor private lemmas, nor removing the Quote dependency, nor virtual-memory caps changed the outcome; ulimit -v is additionally hostile to Chez's address-space reservations. Conclusion: a module-context elaboration pathology, not our code; part S stays a leaf outside the package chain, and the Val datatype it hosted was extracted into the six-line Values module so that nothing downstream needs S.

Finding two, the EyeOne conversion anomaly. Five proof scripts in part T, jTagCase, qMkBeta, readKFires, readKAppSS, readKDeep, fail elaboration in module context with stuck conversions over quote and sub1, while micro-tests prove that exactly those functions reduce perfectly across the same module boundary, and while the same scripts pass in the monolith. Merging Workshop,

Quote, and EyeOne into one module reproduces the same five failures, so the trigger is not the import boundary but something subtler in the regenerated context. The five errors are preserved in the shipped logs as a reproducible artifact; distilling a minimal reproduction is registered as follow-up M1b, a candidate upstream report.

The environment disciplines learned

Swap can be enabled in the sandbox, fallocate, mkswap, swapon, which in principle ends the 4 GB era; but two hazards attach. First, with swap available, Chez's collector turns lazy and balloons, the monolith that fit in under 4 GB under tight GC grew past 13 GB under swap and was killed, so swap is a safety net, not a license. Second, swapfiles eat the small disk; the sandbox filled to 100 percent once, which silently breaks everything, so swap must be sized against df and removed after use. Further disciplines: idris2 —check requires the working directory to match the module name; pkill -f idris2 kills one's own wrapper; detached builds survive tool-call boundaries via setsid but not conversation-turn boundaries, so long builds must be polled within a turn.

Status and next doors

M1 closes at the foundation: the first ten modules are a real package with fast, isolated, repeatable verification, and every future campaign that extends parts A through R can now rebuild only what it touches. M1b is registered for the upper region: minimal reproduction of the two anomalies, then either an upstream issue or a workaround discovered in the distillation. Until then, the monolith remains the verification vehicle for parts T through AC, unchanged in status and honesty.